

Applied Econometrics

Economic and Financial Drivers of Market Fear: An Analysis of the US and the Euro Area

Loucas Di Pillo, Lucas Deweys

s2502123, s2502860

Contents

1	Introduction	3
2	Literature Review	4
3	Theoretical model	4
4	Data	6
4.1	Descriptive statistics	7
4.2	Differences in variables means	9
5	Results	9
5.1	Controlling heteroskedasticity and autocorrelation	12
5.2	Heterogeneous effects between areas	12
5.2.1	Interaction with GDP	13
5.2.2	Interaction with FSI	13
5.2.3	Interaction with EPU	14
5.3	Non-linear model	14
5.3.1	Construction of the high-fear indicator	15
5.3.2	Econometric specification	15
5.3.3	Estimation results	15
5.3.4	Marginal Effects and interpretation	16
6	Discussion and policy recommendation	16
7	Conclusion	17
A	Appendix	20
A.1	Data sources	20
A.2	Data visualization	21
A.3	Logit coefficients	24

List of changes in corrected report:

- Every part was re-written and shortened
- Some parts were deleted because of their small added value and to limit the numbers of pages
- Most abbreviations were replaced by the full name
- Replaced IV (implied volatility) to avoid confusion with instrumental variables
- Table merging
- Some parts and graphs were moved to another section or in the appendix

1 Introduction

Since the early 2000s, big world crises have hit and disrupt financial markets. The subprime, European debt, COVID-19, and energy crises from Russia's invasion of Ukraine were major challenges for the US and Europe. Most crises start with economic and financial issues, but they get much worse as people panicked.

Sometimes, like in the sovereign debt crisis, things got worse because people expected them to, making weak countries even weaker. Sometimes, economic and financial problems spread faster and wider than just the economy's base numbers could explain De Grauwe and Ji (2013). Fear is really the main driver of financial tension. So, grasping what causes this fear and how it shows up in various parts of the economy is really essential for today's market analysis.

This study analyzes dynamics by comparing the United States and the Euro Area. This comparison came about because there are big structural differences between the two regions. On the one hand, the S&P 500, a collection of 500 companies that make up almost 80% of U.S. stock market value, is doing pretty well. The large market capitalization shows a really deep, liquid, and uniform financial market. On the other hand, the Euro Stoxx 50 brings together 50 big companies from many Euroarea countries Buće (2025). These countries have a messy institutional setup and economic heterogeneity De Grauwe and Ji (2013). Making investors worry that during crisis, Europe won't react fast or well, which just increase uncertainty and market volatility. This isn't just a intuition, there's a connection in research looking at how volatility spreads differently (Füss et al. 2011; Badshah 2018).

To capture market fear, this study relies on two implied-volatility indicators: VIX for the United States (based on the S&P 500) and VSTOXX for the euro area (based on the Euro Stoxx 50). Implied volatility shows what the market expects for future price movements, capturing how uncertain people feel. People call these indexes "fear gauges" because they show the worries and thoughts that fill financial markets Ganti (2025).

Even though past studies went deep into what these indicators can predict, they haven't really looked into what actually drives them. This study focus on this, figuring

out if these driving factors are different across economic areas, and what really drives implied volatility, whether macroeconomic fundamentals, financial stress, institutional uncertainty, monetary policy, or exogenous shocks. The next section gives a structured literature review based on these determinants.

2 Literature Review

The literature broadly agrees that implied volatility isn't just about finance, it's a reflection of macroeconomic and institutional environment.

Research shows that macroeconomic fundamentals are important determinants of volatility. Corradi, Distaso, and Mele (2012) point to the business cycle, interest rates, and the labor market as core drivers of volatility. Similarly, Harris and Stoja (2018) argue that both economic activity and investor sentiment are crucial. More recently, Bai and Cai (2024) found that adding macro variables into VIX models actually makes them more predictive, proving that macro fundamentals are essential for gauging risk expectations.

Then, we have the importance of political and institutional uncertainty. Brogaard and Detzel (2015) showed that movements in the Economic Policy Uncertainty (EPU) index lead directly to higher anticipated volatility. This suggests that the VIX captures not just economic facts, but the market's anxiety over political instability.

Moreover, volatility doesn't behave the same everywhere. Badshah (2018) notes that the VSTOXX (Europe) reacts differently to shocks than the VIX (U.S.), likely due to the unique institutional setup of the Euro Area.

Furthermore, central banks exert considerable influence. Lee and Ryu (2019) found that interest rate hikes trigger movements in implied volatility as investors want to adjust their growth and risk expectations quickly.

Finally, outside shocks play a major role. Dupoyet and Shank (2017) highlight how oil price swings, often linked to geopolitical tension, can amplify risk perceptions.

In short, the literature identifies five main drivers of volatility: macro fundamentals, financial stress, institutional uncertainty, monetary policy, and global shocks. While the list of potential variables is massive, it's clear that "fear" is a complex indicator. This study uses these five categories to structure the empirical analysis that follows.

3 Theoretical model

In our study, we consider two periods and two monetary zones. The first period goes from January 2015 to January 2020 and the second one from February 2020 to December 2024. Lockdown measures relative to the COVID-19 were applied in March 2020 in both

zones. Data however suggest that markets anticipated the measures, explaining the choice of February rather than March. We thus use divide in these sub-periods to compare a normal time with a time of crisis. The two regions are the Euro Area and the United States.

The number of variables is deliberately limited to avoid model over-parametrization. Our model relies on five macroeconomic and financial variables. We use the real GDP growth used as an indicator of economic health Corradi, Distaso, and Mele (2012) and Harris and Stoja (2018). Financial stress index is included to capture tensions in financial markets (Office of Financial Research 2025). Economic policy uncertainty index is used to take into account political and geopolitical effects (Baker, Bloom, and Davis 2016). Key interest rates are included to reflect monetary policies and take into account inflation (Lee and Ryu 2019). Oil is the last variable included in our study. We use Brent for the Euro Area and WTI for the United States as oil prices reflect global shocks in the economy Dupoyet and Shank (2017).

No major multicollinearity is found as showed in the table below.

Table 1: Correlation Matrix

	Implied volatility	GDP growth	FSI	EPU	Interest rate	Oil
Implied volatility	1					
GDP growth	-0.171	1				
FSI	0.487	-0.255	1			
EPU	0.367	-0.108	0.483	1		
Interest rate	-0.289	0.0523	-0.0922	0.0826	1	
Oil	-0.141	0.0601	-0.144	-0.0122	0.340	1

We choose to rely on fixed-effect for our model even if our tests suggest otherwise. The assumption that region-specific unobserved heterogeneity is uncorrelated with our explanatory variables is unlikely to hold in this context, given structural economic, financial, and policy differences between the United States and the Euro Area.

In our setting, we estimate implied volatility as follows :

$$\text{Implied Volatility}_{it} = \beta_0 + \beta_1 \text{GDP Growth}_{it} + \beta_2 \text{FSI}_{it} + \beta_3 \text{EPU}_{it} + \beta_4 \text{Interest Rate}_{it} + \beta_5 \text{Oil}_{it} + \alpha_i + \varepsilon_{it} \quad (1)$$

where:

- *Implied Volatility_{it}* is the implied volatility for region *i* at time *t*,
- *GDP growth_{it}* is the real GDP growth rate for region *i* at time *t*,

- FSI_{it} is the Financial Stress Index for region i at time t ,
- EPU_{it} is the Economic Policy Uncertainty index for region i at time t ,
- $Interest\ Rate_{it}$ is the short-term interest rate for region i at time t ,
- Oil_{it} is the oil price for region i at time t ,
- α_i is the region-specific fixed-effect,
- ε_{it} is the error term.

4 Data

We use data from different databases. In the case of the Euro Area, some data are aggregated data to take into account the different country of the monetary zone. Financial stress index and short-term interest rates are published by the federal Reserve Bank of St. Louis for the United States while the ones for the Euro Area is developed by the European central Bank. Data on economic policy uncertainty come from the index developed by Baker, Bloom, and Davis (Baker, Bloom, and Davis 2016). Gross domestic product series for the United States and the Euro Area are provided respectively by the Bureau of Economic Analysis and Eurostat. Oil prices, from WTI for the United States and Brent for the euro area. The S&P 500 and Euro Stoxx 50 stock indices, are obtained from market platforms aggregating official price quotations.

The sample is divided into two distinct sub-periods, which can be described respectively as a period of calm and a period of crisis. The first period goes from January 2015 to January 2020. The second period starts in February 2020 and lasts until December 2024. The analysis is based on monthly data; however, some series are not available at this frequency. Real GDP growth, for example, is observed on a quarterly basis. To ensure data homogeneity, this variable was converted to a monthly frequency by dividing the quarterly value by three and assigning the resulting value to each month of the corresponding quarter. Oil prices and short-term interest rates are initially available at a daily frequency. For these series, monthly averages were computed. An important transformation was applied to the financial stress and policy uncertainty indicators. Because these indicators are constructed using different methodologies and expressed on different scales, they are not directly comparable. To address this issue, the series were standardized using a z-score transformation. This procedure expresses the variables in terms of deviations from their mean, scaled by their standard deviation. The link of all data source is provided in the appendix. Correlations between the variables and implied volatility can also be found in the appendix.

4.1 Descriptive statistics

Here below are the basic statistics of every used variables. Four tables are present, one for every group for every period, that is before and after 2020. Let us first discuss the US values.

Table 2: Descriptive statistics — US (2015–2020)

	mean	sd	min	max
Implied volatility	15.27033	3.895759	9.51	28.43
GDP growth	.810929	.4856352	-1.733333	1.6
Financial Stress Index (std)	.0993617	.6028525	-.8539084	1.675781
Economic Policy Uncertainty (std)	-.3205966	.6315644	-1.188586	1.547926
Interest rate	1.217213	.799056	.25	2.5
Oil price	53.42574	9.111745	33.62	74.15

Table 3: Descriptive statistics — US (2020–2024)

	mean	sd	min	max
Implied volatility	21.75424	8.054933	12.44	53.54
GDP growth	1.007345	3.471549	-9.333333	11.63333
Financial Stress Index (std)	-.1027299	1.286329	-1.136578	7.084299
Economic Policy Uncertainty (std)	.3314642	1.191316	-1.312973	4.588496
Interest rate	2.635593	2.318089	.25	5.5
Oil price	71.77119	19.08841	22.71	114.67

Descriptive statistics reveal a clear regime shift in the U.S. financial environment between the two sub-periods. Over 2015–2020, the VIX displays a relatively low mean (15.3) and moderate dispersion, indicating broadly stable market conditions. From 2020 onward, implied volatility rises markedly, with an average of 21.8 and a substantially higher standard deviation, reflecting increased instability driven by the COVID-19 shock, geopolitical tensions, and monetary policy tightening.

GDP growth also becomes considerably more volatile after 2020, with extreme values ranging from -9.3% to $+11.6\%$, capturing the sharp contraction at the onset of the pandemic and the subsequent rebound. Financial stress follows a similar pattern, shifting from relatively contained levels to a much more erratic dynamic marked by episodes of severe stress. Economic uncertainty increases persistently, as indicated by the transition of the standardized EPU from a negative to a positive mean. Interest rates confirm this regime change, rising sharply after a period of stability, while oil prices increase and become more volatile. Overall, the U.S. data point to a transition toward a highly uncertain macro-financial environment with elevated volatility.

Table 4: Descriptive statistics — Euro Area (2015–2020)

	mean	sd	min	max
Implied volatility	18.74803	5.077476	11.99	32.31
GDP growth	.1360656	.184119	-1.1	.2666667
Financial Stress Index (std)	-.3533152	.4916206	-.9730645	1.270081
Economic Policy Uncertainty (std)	-.4412669	.8062066	-1.606556	2.392672
Interest rate	-.3150457	.1184352	-.5492174	-.036
Oil price	58.3441	11.37312	34.73	82.69

Table 5: Descriptive statistics — Euro Area (2020–2024)

	mean	sd	min	max
Implied volatility	22.63407	7.434576	13.38	48.59
GDP growth	.1322034	1.248232	-3.7	3.833333
Financial Stress Index (std)	.365292	1.238819	-1.024577	3.307891
Economic Policy Uncertainty (std)	.4562251	.9812361	-1.486978	3.070282
Interest rate	1.174578	1.957034	-.5849545	3.907714
Oil price	75.68068	21.34804	22.71	122.84

The Euro Area also experiences a significant increase in volatility and uncertainty after 2020, though the magnitude of fluctuations remains smaller than in the United States. Prior to 2020, the VSTOXX records a moderate average level (18.7), suggesting relatively stable financial conditions despite institutional tensions such as Brexit. After 2020, implied volatility rises substantially, with a mean of 22.6 and greater dispersion, reflecting heightened sensitivity to global shocks, particularly the pandemic and the energy crisis following the war in Ukraine.

GDP growth remains modest on average but becomes more volatile after 2020, with values ranging from -3.7% to $+3.8\%$. Financial stress increases persistently, moving from initially low levels to a positive and more volatile regime, while economic uncertainty rises in a sustained manner, as captured by the standardized EPU. Interest rates, which had been negative for an extended period, increase sharply following the normalization of monetary policy by the ECB. Oil prices also become more volatile, reflecting Europe’s energy dependence. Overall, the euro area shifts from a relatively stable environment to a period of heightened financial stress and uncertainty, albeit less pronounced than in the U.S. case.

4.2 Differences in variables means

A natural follow-up question is whether the computed mean values differ statistically between the United States and the Euro Area. Identifying such differences can help reveal potential structural divergences between the two regions.

To address this, mean-comparison tests were performed for each variable, separately for the non-crisis and crisis periods. During the non-crisis period, the results indicate highly significant differences for implied volatility, GDP growth, the FSI, and oil prices, while EPU levels appear similar across regions. In contrast, during the crisis period, interest rates, GDP growth and the FSI exhibit strong to mild differences between the United States and the Euro Area; the remaining variables display very similar mean values.

January 2015 - January 2020			
	US	Euro Area	Difference (t-test)
Implied volatility	15.27	18.74	-4.24***
GDP growth	0.81	0.13	10.14***
FSI	0.09	-0.35	4.54***
EPU	-0.32	-0.44	0.92
Interest rates	1.21	-0.31	14.81***
Oil	53.42	58.34	-2.63***

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$

February 2020 - December 2024			
	US	Euro Area	Difference (t-test)
Implied volatility	21.75	22.63	-0.61
GDP growth	1.00	0.13	1.81*
FSI	-0.10	0.36	-2.01**
EPU	0.33	0.46	-0.62
Interest rates	2.63	1.17	3.69***
Oil	71.77	75.68	-1.10

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$

5 Results

Table 6 presents our results for the regressions. Our baseline model shows that the main variables are financial stress index, which is also the variable explaining the biggest part

of the model, short-term interest rates and oil prices. We can see that these variables are significant during the normal times, which is prior 2020, and during the crisis, after 2020. The economic uncertainty only appears lightly significant during the normal times. These results confirm that monetary policies (and so inflation), financial matters and exogenous shock (oil) are the most important elements to consider to track implied volatility. The explanatory power of our variables is very different from one to another. In fact, GDP growth only apparently explain 3% of the model, while more than 30% of the model is explained by the financial stress index. Several results should be highlighted. Financial stress, which is the most important variable, is positive in both periods. Interestingly enough, it appears less positive during crisis times while its explanatory power of the model increases. This could potentially mean that financial tensions remain strong while other forms of fear, not taken here into account, might appear alongside. Interest rates also are interesting. We can see that they increase the implied volatility during the calm period whereas they decrease it during the crisis times. This suggests that, in normal times, higher interest rates may tighten financial conditions and increase market uncertainty. The reversal we see in crisis time could be because markets saw the increase in interest rates as a stabilization thereby reducing uncertainty. It is also important to note that interest rates did not evolve in isolation. In the Euro Area, the period from 2015 to 2018 was characterized by substantial net public asset purchases (quantitative easing monetary policy) under the Public Sector Purchase Program (PSPP), followed by the Pandemic Emergency Purchase Program (PEPP) from 2020 to 2022 Van Den End (2024), and a shift toward quantitative tightening after the invasion of Ukraine. These unconventional measures imply that policy rates interacted with a broader set of monetary tools rather than operating in a closed or standalone environment.

A similar situation occurred in the US. After several rounds of quantitative easing following the global financial crisis, the Federal Reserve began raising interest rates in late 2015 while gradually reducing the size of its balance sheet from 2017 onward. With the outbreak of COVID-19, the Fed returned to an accommodative stance, cutting rates to zero and launching massive asset purchases. From 2022 onward, US monetary policy shifted abruptly toward tightening. As in the Euro Area, interest rates in the United States therefore moved within a complex and evolving policy framework shaped by both conventional and unconventional instruments.

Oil price also appears significant and negative, suggesting that an increase in oil price reduce market fear.

Table 6: All regressions

	Baseline model		Robust model		Interaction with FSI		Interaction with EPU		Interaction with GDP	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	2015 - 2020	2020 - 2024	2015 - 2020	2020 - 2024	2015 - 2020	2020 - 2024	2015 - 2020	2020 - 2024	2015 - 2020	2020 - 2024
GDP growth	-1.113 (0.885)	-0.272 (0.190)	-1.113* (0.0959)	-0.272 (0.280)	-1.323* (0.142)	-0.194 (0.276)	-1.107* (0.0941)	-0.289 (0.302)	-1.032 (0.206)	-0.205 (0.155)
Financial Stress Index (std)	4.864*** (0.732)	2.347*** (0.466)	4.864* (0.479)	2.347 (1.581)	4.139*** (0.0222)	3.326 (1.294)	4.837* (0.555)	2.380 (1.589)	4.880* (0.529)	2.332 (1.607)
Economic Policy Uncertainty (std)	-0.845* (0.475)	-0.126 (0.535)	-0.845 (0.526)	-0.126 (1.091)	-0.859 (0.552)	-0.306 (1.091)	-0.649 (0.829)	-0.681 (0.315)	-0.846 (0.529)	-0.123 (1.086)
Interest rate	2.793*** (0.631)	-1.805*** (0.230)	2.793 (1.352)	-1.805 (0.422)	2.498 (1.259)	-1.799 (0.396)	2.713 (1.490)	-1.894 (0.504)	2.802 (1.380)	-1.815 (0.435)
Oil price	-0.0770** (0.0378)	-0.0765*** (0.0246)	-0.0770 (0.0778)	-0.0765 (0.0144)	-0.0684 (0.0865)	-0.0627 (0.0245)	-0.0776 (0.0771)	-0.0870 (0.0190)	-0.0779 (0.0799)	-0.0742 (0.0127)
Euro Area $\times$ Financial Stress Index					1.758 (0.849)	-1.674 (0.872)				
Euro Area $\times$ Economic Policy Uncertainty							-0.288 (0.613)	1.095 (0.840)		
Euro Area $\times$ GDP growth									-0.615 (1.254)	-0.615** (0.0352)
Constant	20.88*** (2.113)	31.16*** (1.884)	20.88 (3.644)	31.16** (1.802)	20.84 (3.969)	30.34** (2.294)	20.95 (3.572)	32.08** (2.427)	20.92 (3.695)	31.02** (1.637)
Observations	122	118	122	118	122	118	122	118	122	118
R^2	0.432	0.592	0.432	0.592	0.441	0.607	0.432	0.596	0.432	0.596

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.1 Controlling heteroskedasticity and autocorrelation

Once heteroskedasticity-robust standard errors are applied and autocorrelation is controlled, most individual coefficients lose statistical significance, particularly during crisis periods. This does not point to a misspecified model; rather, it reflects the inherent difficulty of identifying marginal effects in environments marked by elevated volatility and strong comovements across macro-financial variables.

The relatively high R^2 indicates that the explanatory variables jointly capture broad movements in implied volatility. Yet during crisis episodes, spikes in implied volatility tend to occur simultaneously with shifts in many macro-financial indicators. As a result, while the model remains informative in terms of overall fit, inference on specific parameters becomes less reliable once appropriate corrections are applied.

By contrast, in more stable non-crisis periods, some variables, such as GDP growth and financial stress, retain limited explanatory power, suggesting that observable fundamentals play a more meaningful role outside episodes of systemic turmoil.

5.2 Heterogeneous effects between areas

We now turn to the question of whether differences emerge between the two regions. To do so, we introduce a dummy variable, Euro Area (EA), which takes the value 1 for observations from the Euro Area and 0 for those from the US. Since the objective here is to identify potential differences in the channels through which policies may operate, we restrict interactions to the variables that are plausibly influenced by domestic or regional policy frameworks. Specifically, we interact the EA dummy with GDP growth, the Financial Stress Index (FSI), and the Economic Policy Uncertainty (EPU) index. We do not interact interest rates, whose levels are determined by inflation-targeting central banks policies, nor oil prices, which lie outside the scope of domestic policy.

The purpose of these interactions is to assess whether the impact of these variables differs across regions. A difference in the coefficient on GDP growth would suggest that economic-growth-related policies may have heterogeneous effects across the two areas. A difference in the FSI coefficient would indicate that market-related policies or financial-stability tools may need to be tailored differently. Finally, a difference in the EPU coefficient would imply that communication strategies and policy announcements may have distinct effects in the euro area and the United States, highlighting the importance of how policymakers articulate their intentions.

5.2.1 Interaction with GDP

For the interaction with the GDP, the model is as follows :

$$\begin{aligned} \text{Implied volatility}_{it} = & \beta_0 + \beta_1 GDP_{it} + \beta_2 FSI_{it} + \beta_3 EPU_{it} \\ & + \beta_4 IR_{it} + \beta_5 Oil_{it} + \gamma EA_i \\ & + \delta (GDP_{it} \times EA_i) + \alpha_i + \varepsilon_{it} \end{aligned} \quad (2)$$

The interaction term in table 13 provides insight into regional differences. In the pre-crisis period, the interaction coefficient is negative but statistically insignificant, indicating that the effect of GDP growth on implied volatility does not differ meaningfully between the EA and the US in normal times. During the crisis, by contrast, the interaction term becomes negative and statistically significant at the 5% level. This implies that the EA exhibits a stronger sensitivity of implied volatility to GDP growth during crisis periods. In other words, a given decline in GDP growth increases implied volatility more in the EA than in the US. Equivalently, an improvement in GDP growth reduces volatility more strongly in the euro area during crises. Formally, the total effect of GDP growth for the euro area during the crisis is the sum of the baseline coefficient and the interaction term:

$$\beta_{\text{GDP, EA}} = -0.213 - 0.607 = -0.820$$

This means that, during the crisis, a one-percentage-point increase in GDP growth reduces implied volatility by about 0.82 points in the euro area, compared with only 0.21 points in the United States. This difference suggests that macroeconomic conditions play a more prominent role in shaping volatility expectations in the Euro Area during turbulent periods, possibly reflecting structural differences in market depth, financial integration, or the credibility of policy responses.

5.2.2 Interaction with FSI

For the interaction with the FSI, the model is as follows :

$$\begin{aligned} \text{Implied volatility}_{it} = & \beta_0 + \beta_1 GDP_{it} + \beta_2 FSI_{it} + \beta_3 EPU_{it} \\ & + \beta_4 IR_{it} + \beta_5 Oil_{it} + \gamma EA_i \\ & + \delta (FSI_{it} \times EA_i) + \alpha_i + \varepsilon_{it} \end{aligned} \quad (3)$$

The interaction in table 14 term provides some information about regional differences. In the pre-crisis period, the interaction coefficient is positive (1.914) but statistically insignificant, indicating that the sensitivity of implied volatility to financial stress does

not differ meaningfully between the euro area and the United States in normal times. In other words, before the crisis, a given increase in financial stress raises implied volatility by a similar magnitude in both regions.

During the crisis, however, the interaction term becomes negative (-1.705), although it remains statistically insignificant. While the lack of significance prevents strong conclusions, the sign of the coefficient could potentially suggest that the Euro Area may exhibit a weaker sensitivity of implied volatility to financial stress during crisis periods. In the end, no statistical difference appear between US and EA for the FSI.

5.2.3 Interaction with EPU

For the interaction with the EPU, the model is as follows :

$$\begin{aligned} IV_{it} = & \beta_0 + \beta_1 GDP_{it} + \beta_2 FSI_{it} + \beta_3 EPU_{it} \\ & + \beta_4 IR_{it} + \beta_5 Oil_{it} + \gamma EA_i \\ & + \delta (EPU_{it} \times EA_i) + \alpha_i + \varepsilon_{it} \end{aligned} \quad (4)$$

The interaction term in Table 15 examines whether the effect of EPU on implied volatility differs between the euro area and the United States. The baseline coefficient on EPU is negative in both periods, but statistically insignificant, confirming the earlier finding that EPU does not exert a strong or systematic influence on implied volatility once financial stress, interest rates, and oil prices are controlled for.

The interaction term provides insight into potential regional differences. In the pre-crisis period, the interaction coefficient is small (-0.193) and statistically insignificant. This indicates that, in normal times, the effect of policy uncertainty on implied volatility does not differ meaningfully between the euro area and the United States. In other words, before the crisis, a change in EPU has a similarly weak impact on implied volatility in both regions.

During the crisis, however, the interaction term becomes positive (1.055), but remains statistically insignificant. Here again, the lack of significance prevents firm conclusions, the sign of the coefficient may suggests that the Euro Area may exhibit a stronger sensitivity of implied volatility to policy uncertainty during crisis periods (or may just be noise as it is not significative).

5.3 Non-linear model

Linear models capture average effects on implied volatility, but they do not inform us about the probability of entering a high-fear regime.

To complement regressions, we estimate non-linear probability models that assess how macro-financial variables affect the likelihood of extreme market fear.

5.3.1 Construction of the high-fear indicator

We have build a binary indicator of high market fear, equal to 1 when implied volatility is above the 75th percentile of its pooled distribution across both regions and over the entire sample period, and 0 otherwise.

5.3.2 Econometric specification

We estimate a pooled logit model in which the probability of entering a high-fear regime depends on key macro-financial drivers. The specification includes a region dummy to capture structural differences between the euro area and the United States, as well as year fixed effects to control for common global shocks. Robust standard errors are used.

In addition, to avoid over-parameterization in the non-linear specification, we focus on the main financial and uncertainty-related drivers. Including GDP growth or oil prices does not materially affect the results.

$$\Pr(HighFear_{it} = 1 \mid FSI_{it}, InterestRates_{it}, EPU_{it}) = \Lambda \left(\beta_0 + \beta_1 FSI_{it} + \beta_2 InterestRates_{it} + \beta_3 EPU_{it} + \gamma EA_i + \sum_y \delta_y Year_y \right)$$

where,

$$\Lambda(z) = \frac{1}{1 + e^{-z}}.$$

and, $HighFear_{it}$ is a binary variable equal to one if region i is in a high fear regime at time t . FSI_{it} denotes the Financial Stress Index, $InterestRates_{it}$ the interest rate, and EPU_{it} the Economic Policy Uncertainty index. EA_i is a regional dummy equal to one for the Euro Area and zero for the US. Year fixed effects are captured by $\sum_y \delta_y Year_y$, with one year omitted as the reference category. The pooled logit model is derived from a latent-variable representation in which the error term follows a logistic distribution. This assumption leads to the logistic cumulative distribution function used to model the probability of observing a high fear regime.

5.3.3 Estimation results

Table 16 reports average marginal effects from a pooled logit model estimating the probability of entering a high market fear regime. The estimated marginal effects indicate positive associations between high-fear episodes and both financial stress and economic

policy uncertainty, while the effect of the interest rate is negative. The euro area dummy is positive, suggesting a higher baseline probability of high market fear in the euro area relative to the United States. However, none of these effects are estimated with high statistical precision, indicating that the non-linear results should be interpreted with caution.

5.3.4 Marginal Effects and interpretation

The marginal effects allow for an intuitive interpretation in probability terms. A one-unit increase in the standardized Financial Stress Index is associated with an increase of approximately 4.3 percentage points in the probability of entering a high-fear regime, while a one-unit increase in economic policy uncertainty is associated with an increase of around 2.6 percentage points. Higher interest rates are associated with a lower probability of high market fear, and the euro area exhibits a higher baseline probability than the United States. These effects should be interpreted as suggestive rather than conclusive and are used as a robustness check to the panel analysis.

Table 7: Non-linear estimates: probability of high market fear

	(1)	
	AME	standard error
Financial Stress Index (std)	0.043	0.051
Economic Policy Uncertainty (std)	0.026	0.041
Interest rate	-0.036	0.053
Eura Area (=1)	0.109	0.085
Observations	168	

The corresponding logit coefficients and Pseudo R^2 are reported in Appendix A.

6 Discussion and policy recommendation

Across all our models, financial stress stands out as the most consistent driver of market fear. However, there's a catch: its statistical power is much higher in baseline regressions than in those with robust corrections. We also see a clear split between environments, our models explain about 59% of volatility during crises, but only 43% in calmer times. The takeaway would be, while financial stress is a key player, its impact is sensitive to how you measure it. We should be cautious with the coefficients and remember that "fear" is a complex indicator with many hidden drivers.

Even though some robust regressions show weaker significance, the models still carry weight for policymakers. Since financial stress remains a primary influence, the most

reliable tools for calming markets are macro regulations, liquidity support, and strong oversight. Our findings on interest rates are particularly interesting: In normal times, higher rates tend to push volatility up (likely due to rising borrowing costs). In crisis times, rate hikes actually seem to lower volatility. In the Euro Area, for instance, a rate hike signals that the central bank is finally tackling inflation, which the market views as a stabilizing move. However, while economically, this makes perfect sense. econometrically, these coefficients aren't statistically significant.

Variables like GDP growth, oil prices, and Economic Policy Uncertainty showed much weaker effects. For a policymaker, these aren't really "levers" you pull to fix volatility, they are better used as informational signals.

When comparing the U.S. and the Euro Area, they are more alike than they are different. The only real divergence is that European volatility is slightly more sensitive to GDP drops during a crisis. This suggests that while the two regions react similarly to stress, Europe needs extra policy coordination when the economy get hit.

Finally, our non-linear models confirm that extreme "fear" episodes are almost entirely driven by financial stress and uncertainty. If the goal is to prevent a total market meltdown, policies must target systemic risk and investor confidence above all else.

7 Conclusion

This project examines the determinants of implied volatility by comparing market dynamics in the United States and the Euro Area. The initial model suggests strong effects of financial stress, policy uncertainty, and GDP growth on implied volatility. However, once robust standard errors were applied, most of these effects lost statistical significance, highlighting the importance of careful interpretation. This also show the importance of using robust methods when working with small panel and high-frequency data.

Overall, the results suggest that implied volatility is best stabilized with clear and well communicated financial policies, rather than through adjustment of macroeconomic variables.

This study nonetheless faces several limitations, notably, one of them is the small number of available implied volatility indices, which limit the size of the sample (only two regions). Future research could improve this framework by incorporating more market-based indicators, such as liquidity measures, credit spreads, or broader financial conditions indices, which may better capture the drivers of market fear.

References

- Badshah, Ihsan (2018). “Volatility spillover effects in the European financial markets”. In: *Journal of International Money and Finance* 89, pp. 76–93.
- Bai, Y. and C. X. Cai (2024). “Predicting VIX with adaptive machine learning”. In: *Quantitative Finance* 24.12, pp. 1857–1873. DOI: 10.1080/14697688.2024.2439458.
- Baker, Scott R., Nicholas Bloom, and Steven J. Davis (2016). “Measuring economic policy uncertainty”. In: *The Quarterly Journal of Economics* 131.4, pp. 1593–1636. DOI: 10.1093/qje/qjw024.
- Brogaard, Jonathan and Andrew Detzel (2015). “The asset-pricing implications of government economic policy uncertainty”. In: *Management Science* 61.1, pp. 3–18.
- Buće, Ivan (Dec. 2025). *S&P 500 ou Euro Stoxx 50 ? Le choix stratégique des investisseurs européens en 2026*. FINORUM. URL: https://finorum.com/fr/sp500-ou-euro-stoxx-50-ucits-choix-2026/#FAQ_%E2%80%94_S_P_500_ou_Euro_Stoxx_50_UCITS.
- Corradi, Valentina, Walter Distaso, and Antonio Mele (2012). *Macroeconomic determinants of stock market volatility and volatility risk-premiums*. Tech. rep. 12-18. Swiss Finance Institute.
- De Grauwe, Paul and Yuemei Ji (2013). *The Fragility of Two Monetary Regimes: The European Monetary System and the Eurozone*. Tech. rep. 243. NBB Working Paper. URL: <https://www.econstor.eu/handle/10419/144455>.
- Dupoyet, B. V. and C. A. Shank (2017). “Oil prices implied volatility or direction: which matters more to financial markets?” In: *SSRN Electronic Journal*. DOI: 10.2139/ssrn.2970513.
- Füss, Roland et al. (2011). “The impact of macroeconomic announcements on implied volatility”. In: *Applied Financial Economics* 21.21, pp. 1571–1580. DOI: 10.1080/09603107.2011.583216.
- Ganti, Akhilesh (Dec. 2025). *How implied volatility (IV) works with options and examples*. Investopedia. URL: <https://www.investopedia.com/terms/i/iv.asp>.
- Harris, Richard D. F. and Erdem Stoja (2018). “Macroeconomic uncertainty and the cross-section of stock returns”. In: *Journal of Banking & Finance* 96, pp. 116–131. DOI: 10.1016/j.jbankfin.2018.05.003.
- Lee, J. and D. Ryu (2019). “The impacts of macroeconomic news announcements on intraday implied volatility”. In: *SSRN Electronic Journal*. DOI: 10.2139/ssrn.3313223.
- Office of Financial Research (2025). *OFR Financial Stress Index*. Accessed 2025-12-26. URL: <https://www.financialresearch.gov/financial-stress-index/>.

Van Den End, Jan Willem (2024). “Effects of QE on Sovereign Bond Spreads through the Safe Asset Channel”. In: *International Journal of Finance Economics* 30.2, pp. 1143–1162. DOI: 10.1002/ijfe.2958.

A Appendix

A.1 Data sources

- Financial Stress Index (US): <https://fred.stlouisfed.org/series/STLFSI4>
- Composite Indicator of Systemic Stress (Euro Area): <https://data.ecb.europa.eu/data/datasets/CISS>
- Inflation (US and EA): <https://data-explorer.oecd.org>
- Unemployment rate (US): <https://fred.stlouisfed.org/series/UNRATE>
- Unemployment rate (EU): <https://ec.europa.eu/eurostat>
- Short-term interest rate (US): <https://www.boursorama.com/bourse/taux/cours/historique/2xFEDFNDRT>
- Short-term interest rate (EA): <https://data.ecb.europa.eu/data/datasets/EST>
- Economic Policy Uncertainty Index: <https://www.policyuncertainty.com>
- GDP growth (US): <https://apps.bea.gov/iTable>
- GDP growth (EU): <https://ec.europa.eu/eurostat>
- Oil price (US): <https://fr.investing.com/commodities/crude-oil-historical-data>
- Oil price (EU): <https://www.boursorama.com/bourse/matieres-premier/cours/historique/8xBRN>
- Government debt-to-GDP (US): <https://fred.stlouisfed.org/series/GFDEGDQ188S>
- Government debt-to-GDP (EU): Eurostat (gov_10q_ggdebt)
- Credit to private sector (
- S&P 500 index: <https://www.investing.com/indices/us-spx-500-historical-data>
- Eurostoxx 50 index: <https://www.investing.com/indices/eu-stoxx50-historical-data>

A.2 Data visualization

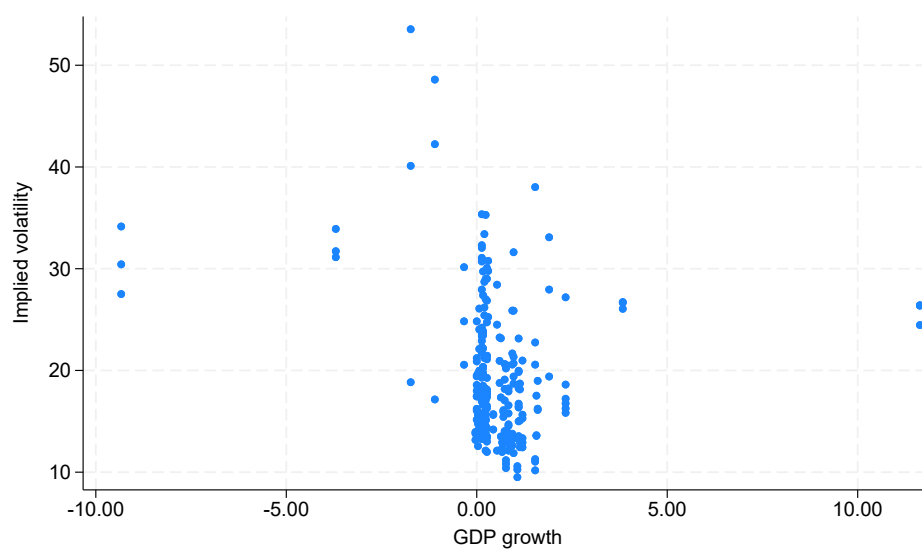


Figure 1: IV vs GDP growth - 2015 to 2020

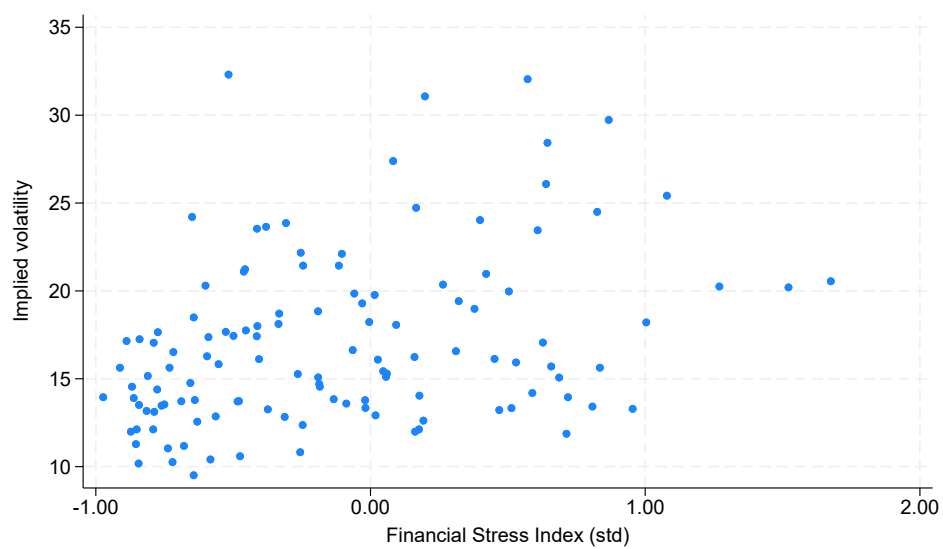


Figure 2: IV vs FSI - 2015 to 2020

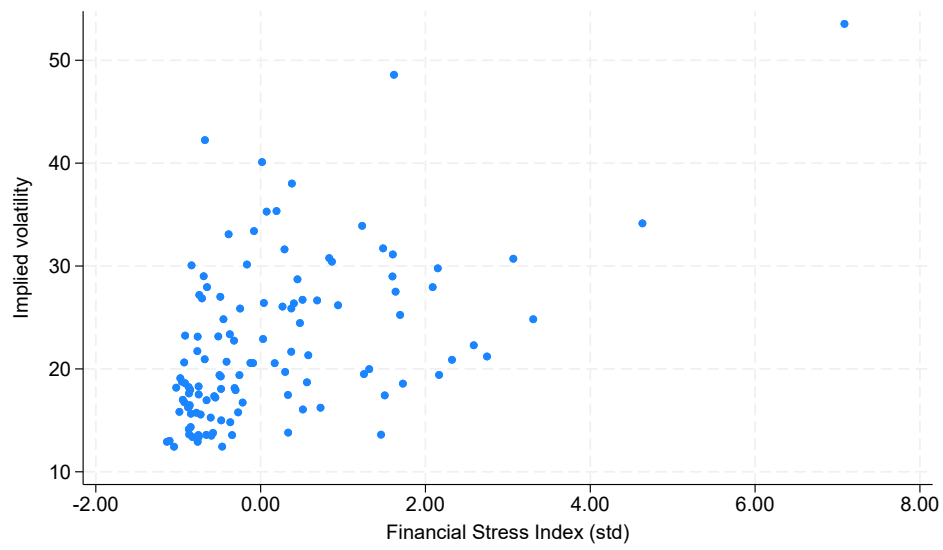


Figure 3: IV vs FSI - 2020 to 2024

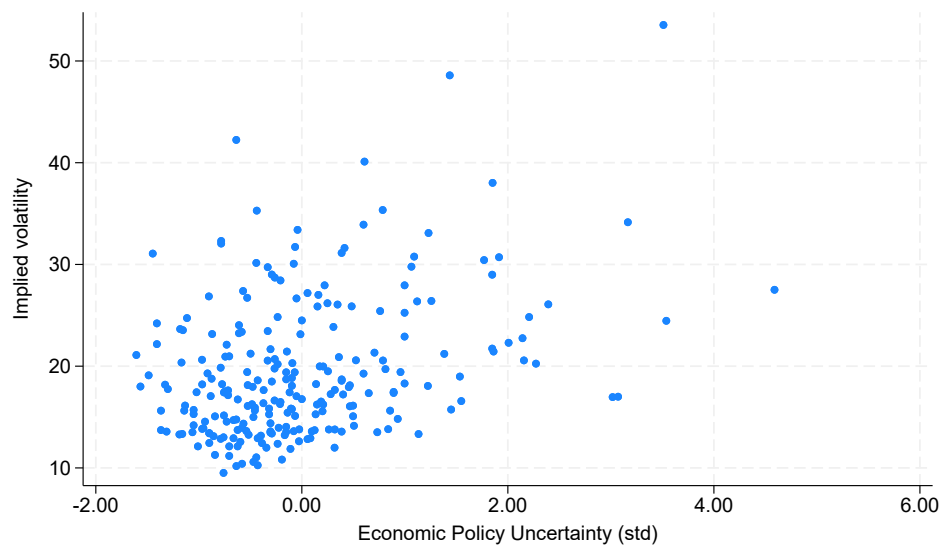


Figure 4: IV vs EPU - 2015 to 2024

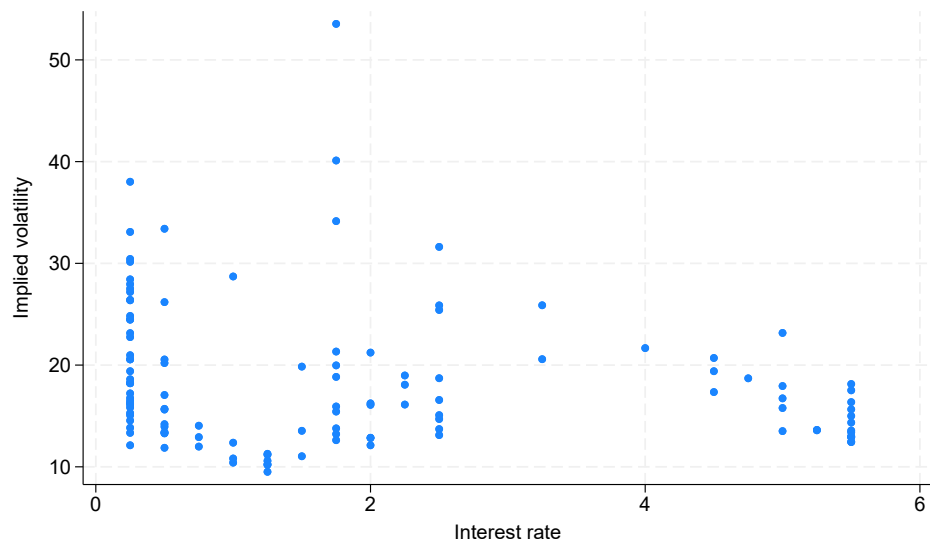


Figure 5: IV vs IR (US) - 2015 to 2020

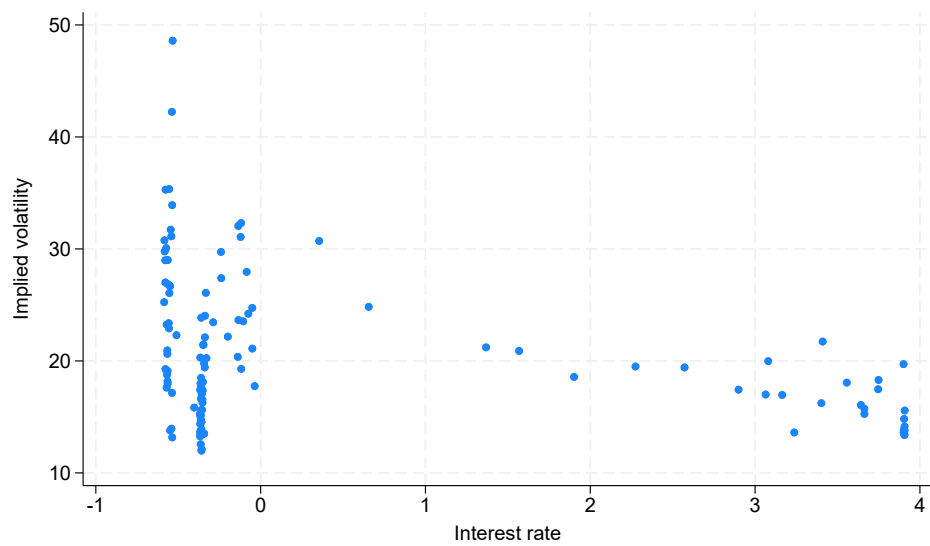


Figure 6: IV vs IR (EA) - 2015 to 2020

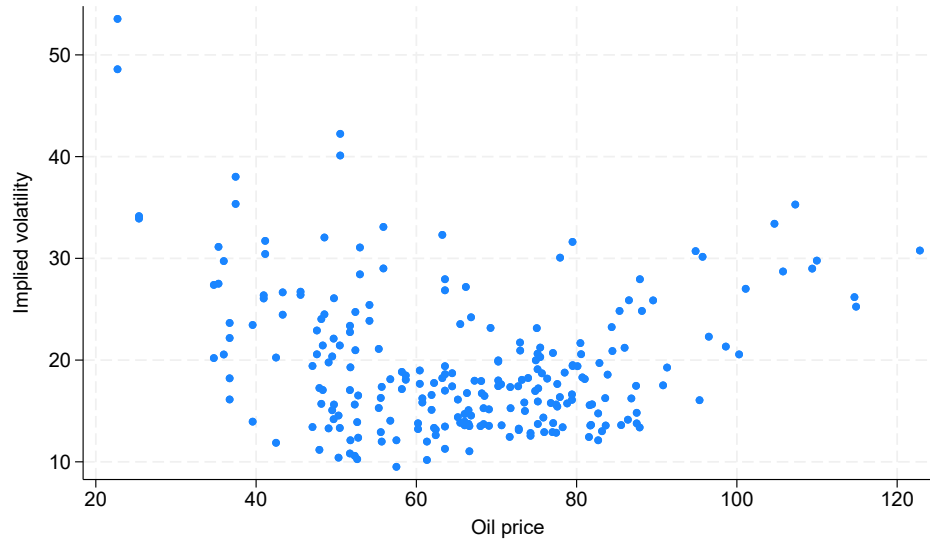


Figure 7: IV vs Oil - 2015 to 2020

A.3 Logit coefficients

Table 8: Appendix A. Logit coefficients

	(1) HighFear
HighFear	
Financial Stress Index (std)	0.295 (0.345)
Economic Policy Uncertainty (std)	0.174 (0.282)
Interest rate	-0.244 (0.364)
Group (EA = 1)	0.743 (0.591)
Calendar year=2015	0 (.)
Calendar year=2016	-1.180* (0.707)
Calendar year=2017	0 (.)
Calendar year=2018	-1.807** (0.876)
Calendar year=2019	0 (.)
Calendar year=2020	1.373 (0.838)
Calendar year=2021	-0.318 (0.731)
Calendar year=2022	1.235 (0.789)
Calendar year=2023	0 (.)
Calendar year=2024	-1.778 (2.234)
Constant	-0.730 (0.562)
Observations	168
Log-likelihood	-76.10
McFadden pseudo- ²	0.305

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$